7.3 The PARC model: understanding policy evaluation

Every access request addresses a fundamental question: Can this principal perform this action on this resource under these conditions? This question forms the basis of Cedar's evaluation model, as well as most modern authorization systems. In the previous section, we explained how policies are structured: Each policy states an effect, defines its scope with a principal, action, and resource, and may include conditions that further restrict when the rule applies.

Those same components also shape the structure of an access request. Cedar formalizes them into a model we'll refer to throughout this chapter and the rest of the book—PARC:

- The *Principal* is the entity making the request, whether it is an employee, a service, or a delegated role.
- The *Action* is what they're trying to do—such as "view," "edit," or "deploy."
- The *Resource* is the object that is being acted upon—a document, system, or record.
- The *Conditions* specify the context of the request—such as time of day, device trust, or geographic location.

While the preceding section focused on policy structure—how rules are written—this section focuses on PARC, which captures how rules are evaluated. The two perspectives mirror each other. A policy's scope maps to the principal, action, and resource in the request, and its conditions are evaluated against the request's context.

In the sections that follow, we will examine how Cedar handles each of these inputs, how matching and evaluation are performed in practice, and what occurs when multiple policies apply to the same request.